Quiz #5

1. The correlation between the heights of husbands and wives in the U.S. is approximately one of the following:

$$-0.9, \qquad -0.3, \qquad 0.3, \qquad 0.9.$$

Which value is most reasonable? Briefly explain your answer.

The correlation is approximately 0.3.

The association should be positive because taller people tend, on average, to have taller spouses. However, the relationship is not especially strong, so a correlation near 0.9 would be too large. Thus, 0.3 is the most reasonable choice.

2. For women age 25 and over in the U.S. in 2005, the relationship between age and educational level (years of schooling completed) can be summarized as follows:

$$\begin{aligned} \text{average age} &\approx 50 \text{ years}, & \text{SD} &\approx 16 \text{ years}, \\ \text{average educational level} &\approx 13.2 \text{ years}, & \text{SD} &\approx 3.0 \text{ years}, \\ r &\approx -0.20. \end{aligned}$$

True or false: As you get older, you become less educated.

Explain your answer.

False.

The correlation of $r \approx -0.20$ shows that, in this population, older women tended to have fewer years of schooling than younger women. It does *not* mean that an individual woman loses education as she gets older.

The negative correlation is largely a cohort effect. Educational attainment increased over time, so younger generations of women generally had more opportunities to complete additional years of schooling than older generations.

Thus, the correlation describes differences *between women of different ages in 2005*, not what happens to the educational level of one woman as she ages.

3. An investigator collected data on the heights and weights of college students. The results are summarized below:

Variable	**Average**	**SD**
Men's height	70 inches	3 inches
Men's weight	144 pounds	21 pounds
Women's height	64 inches	3 inches
Women's weight	120 pounds	21 pounds

The correlation coefficient between height and weight for the men was about 0.60; for the women, it was about the same.

If the men and women are combined into one group, would the correlation between height and weight be

somewhat lower than 0.60, about 0.60, or somewhat higher than 0.60?

Choose one option and explain briefly.

The combined correlation would be somewhat higher than 0.60. Within the group of men, taller students tend to weigh more, giving a correlation of about 0.60. The same is true within the group of women. However, men are also taller and heavier on average than women:

$$70 > 64 \quad \text{and} \quad 144 > 120.$$

Thus, when the two groups are combined, the difference between the group averages reinforces the positive association between height and weight. The combined scatterplot would show a stronger overall positive linear pattern, so the correlation would be somewhat higher than 0.60.

4. Suppose a pandas DataFrame named `students` contains two columns: `study_hours` and `exam_score`. Some rows may be missing one of these two values.

Complete the code below so that the same complete pairs are used to make a scatterplot and compute the Pearson correlation coefficient.

```
pair = students[["study_hours", "exam_score"]]._____()

pair._____(x="study_hours", y="exam_score")

r = pair["study_hours"]._____(pair["exam_score"])
```

(a) Fill in the three blanks.

```
pair = students[["study_hours", "exam_score"]].dropna()

pair.plot.scatter(x="study_hours", y="exam_score")

r = pair["study_hours"].corr(pair["exam_score"])
```

(b) Why is it useful to make the scatterplot before interpreting r?

Correlation summarizes **linear** association. A scatterplot can reveal nonlinear patterns, outliers, clusters, or other structure that a single value of r can hide.

(c) Suppose the code gives $r = 0.68$. Interpret this value in one sentence. Would this value alone establish that studying more causes higher exam scores?

The value $r = 0.68$ indicates a moderately strong positive linear association: students who study more hours tend to have higher exam scores. It does **not** by itself establish causation; other factors and the study design must be considered.